Resources

STAT 109 — Summer Session 1, 2026 (online, async)

Resources

Everything you read and practice lives on this website. Everything you submit for a grade lives in Canvas.

Start here

Resource	Link	When to use
Course path	Schedule	readings, labs, quizzes, projects
Syllabus	Syllabus	Policies, grading, AI expectations

Reference sheets

Sheet	Link	Purpose
Methods map	Methods	When to use which test or summary; study-design vocabulary
R reference	R reference	Code patterns and syntax

R and notebooks

This async course runs R in Google Colab. Lab and project pages link to notebooks on the public GitHub repo; open a link and Save a copy to Drive before editing.

Step	What to do
1	Open the Colab link from a demo lab or exercise lab
2	**File → Save a copy in Drive** (or download `.ipynb` and upload to Canvas)
3	Run cells top to bottom; fix errors before submitting
4	Submit the completed `.ipynb` in Canvas

Colab home: colab.research.google.com (remember to change runtype to R as Python 3 is the default for new notebooks)

Colab tutorial: Google Colab overview — running cells, saving a copy to Drive, editing notebooks. Keep in mind that Colab can be used for multiple programming languages but the basic use apply to how we'll use it to program in the R language.

Optional: R on your own computer

You do not need a local install for this course. If you prefer RStudio or Jupyter on your machine:

- **R:** CRAN · Windows/Mac/Linux installers
- **RStudio (Posit):** RStudio Desktop — install R first
- **R in Jupyter:** IRkernel setup

R is also installed on campus computers (ROSE may have free used laptops if you need hardware).

Quarto and reproducible reports

Projects and some readings use **Quarto** (`.qmd`) — text + code that renders to HTML or PDF. Optional if you want reproducible write-ups beyond plain notebooks:

- Quarto documentation
- Posit cheatsheets — R, `{tidyverse}`, `{ggplot2}`, and more

Render locally: `quarto render your-file.qmd` (requires Quarto and R).

Textbook (optional)

Introductory Statistics for the Life and Biomedical Sciences by Julie Vu and David Harrington — free PDF at OpenIntro or Leanpub. Module reads on this site have all the essential topics but it can be useful to see alternative explanations in someone else's words.

Note: The textbook ships with companion R labs written in base R. This course uses Google Colab, `{tidyverse}`, and `{ggplot2}` in our own lab notebooks.

Practice quiz banks

Open-ended practice questions for each quiz. Canvas quizzes use a random autograded subset.

Module	Quiz A	Quiz B	Synthesis
0	Start Here Check	Question Ideas	—
1	Def	Concept	Synthesis
2	Def	Concept	Synthesis
3	Def	Concept	Synthesis
4	Def	Concept	Synthesis
5	Def	Concept	Synthesis

Optional image-based graph interpretation practice: Graph Interpretation Pack.

Optional Supplemental Materials by module

Module	Textbook (Vu & Harrington)	Videos & tutorials
0	Preface / book home	Colab tutorial · Posit: Getting Started with R · Navarro: R for beginners
1	Ch 2 Probability · Ch 3 Distributions (§3.2 binomial)	Khan: binomial distribution · Khan: visualizing binomial
2	Ch 3.3 Normal · Ch 4 Foundations (§4.1–4.2)	Khan: confidence intervals · Khan: CI simulation · Mine Çetinkaya-Rundel (search: confidence intervals)
3	Ch 1 Intro to data (§1.4–1.7)	R4DS: Data visualization · Posit: ggplot2 video
4	Ch 5 Inference for numerical data (§5.3) · Ch 8 Inference for categorical data (§8.1–8.2)	R4DS: Data transformation · Andrew Heiss (workflow, comparing groups)
5	Ch 6 Simple linear regression	Khan: regression line · Penn State: prediction intervals · Regression shuffle applet

Site-wide companions

Resource	Link	Best for
R for Data Science	r4ds.hadley.nz	`{tidyverse}` + `{ggplot2}` (Modules 3–5)
Posit cheatsheets	posit.co/resources/cheat-sheets	Building your R reference
Posit videos	posit.co/resources/videos	Beginner R workflow
Mine Çetinkaya-Rundel	youtube.com/@minecr	Statistics-first explanations

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Getting help

- **Email:** rho3@humboldt.edu (Cal Poly Humboldt account; allow one business day during the session)
- **Office hours:** By appointment and in Canvas Discussion
- **Canvas:** Announcements, grades, and module discussions

Campus support

- **Computer labs:** R installed on campus machines
- **ROSE:** Reusable Office Supply Exchange — free used computers when available
- **SDRC:** Student Disability Resource Center — accommodations (email instructor your letter of accommodation)
